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Review

Do relationship education programs reduce relationship aggression? A meta-analytic study

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ABSTRACT

There is an increasing focus on evaluating the effectiveness of Relationship Education (RE) programs on reducing relationship aggression. Nevertheless, there has been little by way of a systematic quantitative synthesis of research to date. The primary aim of this research was to conduct a meta-analysis into the effects of RE programs on relationship aggression and provide a comprehensive assessment as to the moderating effects of various methodological characteristics of studies. A secondary aim was to determine whether RE programs reduce negative aspects of relationship functioning that are known to exacerbate relationship aggression. Thirty-one studies (n = 25,527) were included comprising of pre-post comparison studies and control trials. Overall, RE programs were significantly associated with reductions in relationship aggression (d = 0.11, p = .001). Pre-post studies yielded a significantly larger effect size (d = 0.28, p < .001) than RCT studies (d = 0.05, p = .10). Subgroup analysis revealed that participants who reported moderate to severe relationship aggression upon RE program entry demonstrated large reductions in physical (d = 0.66, p = .01) and psychological (d = 0.85, p < .001) aggression compared to those who reported minimal to low relationship aggression on entry (physical aggression d = 0.07, p = .009; psychological aggression d = −0.04; p = .17). Amongst participants who reported moderate to severe relationship aggression, RE programs were also found to reduce controlling behavior (d = 0.20, p < .01) and conflict behavior (d = 0.40, p < .001). Findings demonstrate the emerging efficacy of RE programs for reducing relationship aggression and conflict behavior, particularly in those with a history of moderate to severe levels of relationship aggression.

1. Introduction

Programs developed to address aggression within romantic relationships (i.e., the perpetration of physical and psychological abuse, and the use of coercive controlling behaviors against a romantic partner; WHO, 2013) have been largely informed by a feminist analysis of relationship aggression (e.g., Dobash and Dobash, 1979). Often labelled as Batter Intervention Programs (BIPs), these programs are most commonly delivered to male perpetrators of couple violence who are referred into such programs through the justice system (see Eckhardt et al., 2013). However, over the last decade, there has been an increased awareness that individuals or couples who enter relationship education (RE) programs report ongoing relationship aggression or are at greater risk of aggression in their relationship (Bradford, Skogrand, and Higginbotham, 2011; Hawkins, 2011). An increasing number of these studies have begun to measure changes in relationship aggression as a potential outcome of RE. This raises two important and unanswered questions. (1) does participating in RE mitigate the perpetration of relationship aggression? 2) for those individuals and couples that report relationship aggression, do the benefits, if any, of RE extend beyond reducing aggression to also attenuate negative aspects of relationship functioning (e.g., destructive conflict patterns) that are known to exacerbate the perpetration of relationship aggression (Finkel and Eckhardt, 2013)? To address these questions, this paper reports on a meta-analysis into the effects of RE programs within the context of relationship aggression.

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2. RE and relationship aggression

Studies into the effects of RE programs within the context of relationship aggression have been largely undertaken across two lines of research. One line has specifically focused on evaluating the effects of relationship education on reducing relationship aggression in samples of individuals or couples that report existing relationship aggression (e.g., Cleary Bradley and Gottman, 2012). Another line has focused on evaluating the effects of relationship education programs for low-income couples (e.g., Charles, Jones, and Guo, 2014). This is because low household income is a critical demographic risk factor of experiencing relationship aggression in romantic relationships (e.g., Gustafsson et al., 2016) – a finding that is consistent across ethnic groups (Cunradi, Caetano, and Schafer, 2002). This second line of RE research (with a focus on low-income couples) stems largely from the Healthy Marriage and Relationship Education Initiative launched by the United States Government in 2002 (Administration of Children and Families [ACF, 2002]). The initiative entailed the publicly funded roll-out of RE to socially disadvantaged couples with the aim of enhancing the personal and relationship wellbeing of families through fostering more positive, stable, and nurturing relationships in the family home. This nation-wide roll-out was subject to an extensive evaluation that commenced in 2006 and entailed evaluation data spanning 30 months (Dion and Hawkins, 2008; Lundquist et al., 2014).

When it comes to relationship aggression, both lines of research into RE assume that the relational milieu is an important proximal factor in how couple interactions escalate into acts of relational aggression. This assumption also aligns with an interdependence theory perspective on close relationships (Kelley et al., 1983) as well theory and research into conflict patterns and interpersonal aggression (Christensen, Doss, and Jacobson, 2020; Epstein, Werlinich, and LaTaillade, 2015). From an interdependence theory perspective, romantic relationships are typically characterized by a degree of interdependence, such that the actions of one dyad member not only affect their own behaviors and outcomes, but a partner's behaviors and outcomes (Kelley and Thibaut, 1978; Rusbult and Van Lange, 2003). These behaviors and outcomes can indeed extend to acts of relationship aggression (VanderDrift, Loerger, and Arriaga, 2019). From a couple conflict perspective (e.g., Gottman, 1998; Hellmuth and McNulty, 2008) and research into interpersonal models of relationship aggression (i.e., impelling-instigating-inhibiting [I-cubed] model of aggression, Chester and DeWall, 2018; Finkel and Eckhardt, 2013); an escalating cycle of destructive conflict behaviors entailing criticism, contempt, hostility, and ridicule, known as negative reciprocity, is one of the strongest predictors of relationship aggression (O'Leary, Slep, and O'Leary, 2007). These perspectives highlight that investigating the effectiveness of RE within the context of relationship aggression should not solely focus on whether RE results in reductions in the perpetration of aggression. Moreover, the focus should also extend to whether couples that experience aggression and participate in RE programs report reductions in negative relationship processes and attitudes that can be pre-cursors of relationship aggression.

From a RE vantage point, relationship processes and attitudes reflect modifiable factors, so that the building of awareness, education, and relationship skills can result in the attenuation of destructive interaction patterns that can otherwise escalate into relationship aggression. It is therefore not uncommon for RE programs to comprise of four broad components: awareness, feedback, cognitive change, and skills training (Halford, Markman, Kling, and Stanley, 2003; Hawkins, Carroll, Doherty, and Willoughby, 2004). *The awareness component* focuses on the transmission of information, clarification of expectations, and increasing couples' awareness of key relationship processes that influence relationship outcomes. For example, most programs assist couples to become aware of common problems to avoid. Awareness of basic problems helps to motivate couples to work harder on their relationships (Braithwaite and Fincham, 2014). In addition, virtually all programs discuss or imply that maintaining healthy romantic relationships requires sustained effort. The *feedback* component typically includes the administration of a self-report survey that assesses a variety of personal factors as well as relationship domains such as [but not limited to] commitment, satisfaction, forgiveness, and communication patterns [e.g., PREPARE; Fowers, Montel, and Olson, 1996; RELATE; Larson et al., 1995]). Each member of the couple completes the assessment independently and the couple is provided with structured feedback and interpretation regarding relationship functioning. *Cognitive change* attempts to encourage cognitions believed to promote positive couple relationships. For example, Markman, Stanley, and Blumberg (2001) advocate for discussions that promotes commitment to the relationship and benign (non-blaming) attribution of the causes of negative partner behavior. The *skills training* component entails lecture-based content with situational videos and demonstrations about relationship skills (such as developing constructive patterns of communication and conflict, the positive expression of affection, empathic accuracy, and alike). Many RE programs also include an opportunity to practice these skills and receive feedback from program educators/facilitators.

Despite the broad components that are common across many programs, and the underlying rationale and potential of RE to address relationship aggression, the evidence is somewhat mixed regarding the extent to which RE programs result in reductions in relationship aggression. For example, Antle, Karam, Christensen, Barbee, and Sar (2011) found that low-income individuals reported significant reductions in both physical and emotional abuse at six months post-program completion. In contrast, Wood, McConnell, Moore, Clarkwest, and Hsueh (2010) found that as part of a large-scale Randomized Control Trial (RCT) of an RE program delivered to couples transitioning into first-time parenthood, that participants demonstrated no significant changes in the reported levels of relationship aggression. There are also studies that find RE yields positive outcomes for some aspects of relationship aggression and not others. For instance, Owen, Antle, and Quirk (2017) found that at six-month follow-up, participants that undertook a 16-hour RE program over four weeks reported significant reductions in controlling behaviors, but no significant reductions in physical abuse compared to a control condition.

In the only meta-analysis to date on the effects of RE programs on relationship outcomes that included relationship aggression, Hawkins and Erickson (2015) found a small positive and significant effect for RE programs on relationship aggression. However, the effectiveness of RE programs was limited to control group studies ($d = 0.06$), with no significant effects for one-group/pre-post studies ($d = 0.02$). Although the Hawkins and Erickson (2015) meta-analysis included relationship aggression as an outcome, this was not the exclusive focus of their quantitative synthesis – rather – the focus was on the effectiveness of RE programs for low-income couples and entailed other outcomes such as relationship quality and communication skills. To this end, no analysis was provided as to whether RE programs were more or less effective for different types of abuse (i.e., physical, emotional), and for participants that varied in their level of relationship aggression upon entry into the RE programs. As a case in point, it is plausible that participants who experience higher levels of relationship aggression may experience greater gains by taking part in RE than participants, who at the outset, evidence little by way of relationship aggression. However, it is also plausible to assume that individuals and couples who report severe relationship aggression may experience lesser gains because it may be more difficult to shift problematic relationship dynamics for those with embedded and chronic patterns of aggression. Thus, a quantitative synthesis of research into the effects of RE on relationship aggression that examines different types of relationship aggression and differences in participants existing levels of relationship aggression are critical. Such a synthesis can inform the future targeting and implementation of RE programs in potentially addressing couple and family violence and aggression.

Moreover, any quantitative synthesis of the research into the effects of RE on relationship aggression should (where possible) systematically investigate whether the findings to date may be explained by the methodological characteristics of evaluation studies. Examples of such methodological characteristics include (but are not limited to): program implementation and delivery, study design (e.g., one group pre-post evaluations, RCTs), and the populations in which programs are trialed. A systematic investigation of moderators can provide important insights as to whether the somewhat mixed findings regarding the effects of RE on relationship aggression outlined above reflect systematic variation that can be attributed to the design components of specific studies, the types of RE interventions employed, the socio-demographic profile of couples, and, importantly, levels of aggression upon program entry. Understanding the role of such factors can help to understand the types of RE programs that bring about change as well as the individuals and couples that most (as well as least) benefit from RE when it comes to addressing relationship aggression.

The merit of investigating methodological aspects of RE research (e.g., study design characteristics, sample characteristics, and program elements [i.e., program hours, delivery method]) is highlighted by a previous meta-analysis by Hawkins, Stanley, Blanchard, and Albright (2012) into the programmatic moderators of RE program effectiveness on communication skills and relationship quality. Hawkins et al. identified considerable heterogeneity in the effect sizes as a function of a wide range of factors including (but not limited to): research design (pre-post study designs vs control-group studies), program content, method and setting of delivery, and the number of hours comprising the program. Overall, Hawkins and colleagues found that program dosage and emphasis of programs on communication skills were important moderators of relationship satisfaction, and an emphasis on communication skills was associated with a larger positive effect for couple communication. Furthermore, studies employing a one-group pre-post research design yielded larger effect sizes for relationship satisfaction and communication skills than control-group studies. Findings across meta-analyses regarding the effectiveness of RE programs for participants from a middle-class background (Hawkins et al., 2004) or participants experiencing social and economic disadvantage (Hawkins and Erickson, 2015), suggest that low-income couples report small (if any) gains in terms of communication skills and relationship quality (Hawkins and Erickson, 2015), whereas couples from moderate- and high-income brackets experience gains of moderate effect size (e.g., Hawkins et al., 2004). Even though these previous quantitative reviews point to specific methodological differences and socio-demographic characteristics that are important factors to consider when interpreting the evidence to date, there exists no quantitative synthesis that examines these methodological factors in relation to the effects of RE programs on relationship aggression.

3. Research aims

The primary aim of this research was to conduct a meta-analysis into the effects of RE on relationship aggression and provide a comprehensive assessment as to moderating effects of various methodological characteristics of studies. The secondary aim was to determine whether RE programs reduce negative aspects of relationship that are known to exacerbate relationship aggression. In relation to the first aim, we derived a series of research questions to frame the current meta-analysis. First, do RE programs result in reductions in overall relationship aggression? Second, do RE programs demonstrate reductions across different types of relationship aggression? Third, do RE programs result in greater reductions in relationship aggression for those who report higher (compared to lower) relationship aggression at program entry? Fourth, to what extent do methodological characteristics of evaluation studies (i.e., program and delivery, dosage response, study design, and sample characteristics such as gender, socioeconomic status and education) moderate the effectiveness of RE programs on relationship aggression? In relation to the second aim, our research question was, if RE programs are designed to reduce negative relational behaviors and processes, would those who experience higher levels (compared to lower levels) of relationship aggression upon program entry experience decreases in negative aspects of relationship functioning?

4. Method

A systematic literature search was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA; Page et al., 2021) guidelines. The eligibility criteria, search strategy and process for study selection, quality assessment, and data extraction are described below.

4.1. Eligibility criteria

4.1.1. Program

Eligible studies included those that reported on an evaluation of a relationship education program (including parenting programs with an emphasis on relationship education) with an assessment of relationship aggression. Programs included had to deliver relationship education to individuals or couples. For the purpose of this meta-analysis, relationship education is defined as programs that entail skills-based training and psychoeducation to enhance relationship functioning. We specifically did not include interventions that delivered relationship/couples therapy (e.g., behavioral couples therapy, integrative behavioral couples therapy, emotionally focused couples therapy) as this is distinct from relationship education. Unlike relationship education, which has an emphasis on delivering lecture-based content along with skill building activities to enhance relationship functioning, couples therapy typically includes very limited psychoeducation (though skill-building is a feature of some therapies). Furthermore, various forms of couples therapy place emphasis on working with the cognitions and emotions of each partner to unpack their triggers and sensitivities that contribute to their relationship problems (e.g., Christensen et al., 2020). In addition, couple therapy focuses on addressing each partner's historical factors that can affect relationship functioning (e.g., Halford et al., 2003; Markman and Ritchie, 2015). Couple therapy also involves the extensive assessment of couples for the development of case formulation and treatment planning (e.g., Halford et al., 2003; Markman and Ritchie, 2015). Therefore, the differences in the focus, delivery and format of relationship education and couples therapy warrant the exclusion of therapy studies.

4.1.2. Study design

Studies were included if they compared the intervention to a randomized or non-randomized control group, or if they included a comparison of the sample pre-post intervention.

4.1.3. Types of participants

Participants were adult individuals, couples or parents who were enrolled in relationship education programs (including parenting programs with an emphasis on relationship education) and had been administered an assessment of relationship aggression prior to commencing the study. Where possible, the mean age and age range of the sample as well as the gender composition are reported.

4.1.4. Types of outcomes

RE studies were included if they reported on any form of relationship aggression such as the experience of physical, emotional, or psychological aggression or abuse, and controlling behavior. For those studies that reported on any assessment of relationship aggression, additional outcomes were included if the study included an assessment of negative relationship functioning (e.g., destructive conflict behavior, lack of intimacy, partner distrust).

4.1.5. Length of follow-up

Studies were not required to report a follow-up period to be included in the current review, however this data is reported where it was available.

4.1.6. Publication status

The current review included peer-reviewed studies and grey literature, such as technical reports in English only.

4.2. Search strategy

Parallel database searches for published and unpublished studies were conducted in PsycINFO, MEDLINE Complete, CINAHL Complete, Embase, Scopus, and ProQuest Complete using key terms relating to relationship education evaluation and relationship aggression (see Table S1 in supplemental material for detail). In addition to this, the reference lists of all included articles were also hand-searched, and grey literature searches for technical reports, known programs, and known authors in the field were also conducted via Google Scholar. Initial searches were conducted during April 2019 and an updated search of the databases was conducted in August 2022. All references were imported into Rayyan, a systematic review screening application (Ouzzani, Hammady, Fedorowicz, and Elmagarmid, 2016), for screening.

4.3. Data extraction process and management

Data extracted from the studies included details about the sample, level of aggression/violence at baseline, study design, intervention type, age, gender, comparison group details, retention and response rate, number of program hours, number of assessments and point at which these occurred. Data for all relevant outcome variables (i.e., forms of relationship aggression and aspects of negative relationship functioning) were also extracted.

4.4. Quality assessment and risk of bias

To assess the quality of the studies included in the current review, we utilized the Cambridge Quality Checklists (Murray, Farrington, and Eisner, 2009) (see supplemental material Table S2). The quality of each study was measured across three domains, with points allocated where certain criteria were met for each of these domains, resulting in three separate scores (i.e., one score per domain) for each study. The first domain termed "correlate score" contains five items assessing the study's sampling method, response rate, sample size, and quality of measures used to determine how correlates and outcomes are assessed. Each item is rated either 1 (*indicating total population sampling, adequate response rate, adequate sample size, and reliable/valid measures*), or 0 (*indicating convenience sampling, low response/retention rate, small sample size, and unreliable/invalid measures*). These five items are then summed to yield a total score out of 5 for the first domain. The second domain termed "risk factor score" assesses study design regarding the risk factor of interest, and includes a single item, scored on a 3-point scale, 1 (*cross-sectional design*), 2 (*retrospective design*), and 3 (*prospective design*). The third domain termed "causal risk factor score" assesses a study's ability to make causal inferences regarding change in risk factor (s) over time and includes a single question, scored on a 7-point scale, ranging from 1 (*study without a comparison group and no analysis of change*) to 7 (*randomized experiment with analysis of change*).

Risk of bias was assessed using the Risk of Bias in Non-randomized Studies of Interventions (ROBINS-I; Sterne et al., 2016). The risk of bias assessment was conducted to assess the appropriateness of combining non-randomized studies (i.e., pre-post studies and non-random control studies [seven in total]) as part of an overall quantitative synthesis (see supplemental material Table S3). The ROBINS-I assesses seven domains of potential bias. These domains are: (1) bias due to confounding; (2) bias in selection of participants into the study, (3) bias in classification of interventions, (4) bias due to deviations from intended interventions, (5) bias due to missing data, (6) bias in measurement of outcomes, and (7) bias in selection of the reported result. Each domain of bias consists between three to seven signalling questions that aim to facilitate judgment regarding the risk of bias. Each question is assessed on the basis of 5 response options: (1) yes, (2) probably yes, (3) probably no, (4) no, and (5) no information. A risk of bias judgment is derived for each domain, which can then be used to arrive at an overall judgment of bias. The domain-level and overall level judgment with regards to the risk of bias ranges between low risk of bias to critical risk of bias. Studies (especially those non-randomized study interventions [such as pre-post study designs]) can be combined when the overall risk of bias is judged as low to moderate (Sterne et al., 2016). Studies that are judged to be of serious risk of bias (this overall judgment is reached even if bias is assessed as serious in only one domain), may only be combined providing aspects of study design are homogeneous to the other studies in various respects (i.e., the intervention being trialed, sampling population, outcome measures, etc.). However, such studies should be included with caution, and subgroup analyses and sensitivity analyses can play an important role in further understanding the risk of bias (Sterne et al., 2016). Any study judged as being critical risk of bias is excluded from synthesized analyses (Sterne et al., 2016).

4.5. Publication bias

Funnel plots of the main combined effects for relationship aggression and negative aspects of relationship functioning were created and visually inspected for the potential influence of publication bias on the meta-analyses. Asymmetrical distributions of studies around the averaged effect size indicates the potential for the presence of publication bias. Effect size distribution was then formally tested using Egger's test of the intercept (Egger, Smith, Schneider, and Minder, 1997). A significant Egger's test of the intercept indicates the presence of significant publication bias.

4.6. Sensitivity analyses

To determine the stability of the average effect size estimates, one-study-removed analyses were conducted separately for relationship aggression and negative relationship functioning outcomes in CMA. The one-study-removed analysis runs a series of meta-analyses, where each analysis in the sequence removes one primary study and recalculates the averaged effect (Borenstein, Hedges, Higgins, and Rothstein, 2009). Studies are deemed an outlier if the re-calculated effect size with the study removed falls outside the 95% confidence interval of the overall average effect size estimate. Where outliers were detected, we used a z-test to estimate the statistical significance of the difference between the estimated effects with the outlier(s) included versus with the outlier(s) removed.

4.7. Statistical analysis

4.7.1. Calculation of effect sizes

The meta-analysis was conducted in Comprehensive Meta-Analysis version 3.3.070 (CMA; Biostat Inc., Englewood, NJ; Borenstein et al., 2009). Extracted effect sizes from the included primary studies were entered into CMA. However, given effect size metrics varied across primary studies, effect sizes were converted to a common metric – Cohen's d – to allow for cross-study comparisons. To account for heterogeneity between studies, a random effects model with restricted maximum likelihood estimation was used for all statistical analyses (Borenstein et al., 2009; Card, 2012).

The analyses were conducted in three stages. In the first stage, the effect size for standardised mean difference (d) between relationship education program interventions and all forms of relationship aggression combined (i.e., physical aggression, psychological aggression/emotional abuse, and controlling behavior) was estimated, as were the estimates for each form of aggression separately. In the second stage, we estimated the effect size for relationship education programs and all negative aspects of relationship functioning (e.g., destructive conflict behavior) as well as on each negative aspect separately. Although some studies reported data on outcomes across multiple time points, we used the last timepoint reported in any given study in deriving effect size estimates. In the third stage, subgroup analyses were conducted, which are described below.

4.7.2. Subgroup analyses

A series of subgroup analyses were performed to determine whether the effect of relationship education on each outcome differed as a function of three categories of methodological characteristics: (1) sample characteristics (none-to-low aggression reported at baseline versus moderate-to-severe aggression, gender, SES), (2) program characteristics (method of program delivery, dose-response relationship, program), or (3) study characteristics (study design, follow-up period). Subgroup analyses are conducted where a subgroup combined two or more studies and the sample size, effect size, and degree of heterogeneity, were such that the power could be estimated at a minimum 0.80 (Griffn, 2021). *Z*-tests were used to compare effect size estimates for each subgroup analysis. If there were no statistically significant differences between the subgroup effect estimates, it would suggest the characteristic did not moderate the effect of relationship education programs on the outcome.

4.7.3. Testing heterogeneity

Heterogeneity of the averaged effect sizes was assessed using the Q statistic (Cochran, 1954) and the I^2 (Higgins and Thompson, 2002). A significant Q statistic signifies the presence of high heterogeneity and suggests there is greater variation in the observed effect than would be expected by sampling error alone (Huedo-Medina, Sánchez-Meca, Marín-Martínez, and Botella, 2006). I^2 is an index that reflects the ratio of true heterogeneity to the total variance across observed effect size estimates (Borenstein et al., 2009), where values range from 0 to 100 and higher values indicate greater heterogeneity.

5. Results

5.1. Study selection

Fig. 1 shows the PRISMA flowchart, documenting studies that were included in the current review. A total of 818 records were identified from the searches. After removing all duplicates, one reviewer (AC) screened all titles and abstracts for inclusion (n = 755). A random sample of the identified records (n = 102) were double coded by a research assistant who acted as a second coder. Coders achieved a 98% agreement rate, and where they were unable to agree, articles were

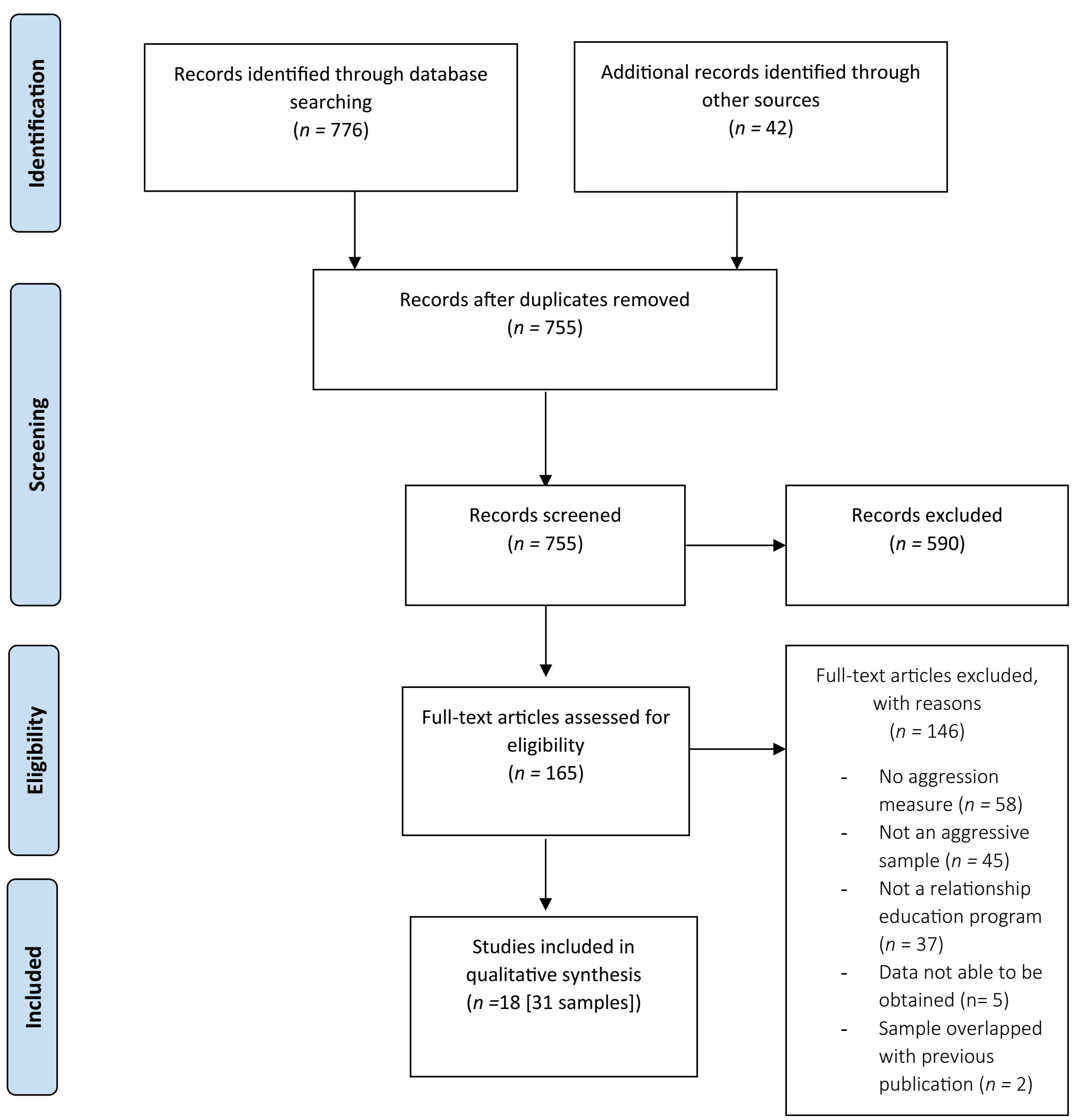

Fig. 1. PRISMA diagram.

discussed with the remainder of the authors to determine if the study met the inclusion criteria. This resulted in 18 records being included in the review. Of these 18 records, two detailed multiple studies that were part of two large evaluation programs. The first of these records reported multiple studies ($k = 8$) that assessed the effect of relationship education programs delivered as part of the Supporting Healthy Marriages (SHM) initiative using eight independent samples across the United States. Similarly, the second of these records detailed multiple studies ($k = 8$) assessing the effect of relationship education programs delivered as part of the Building Strong Families (BSF) Project using independent samples, also across the United States. We chose to include each of the studies comprising the two large evaluations as separate study entries for two reasons. First, each evaluation reported on the details (including outcome data) for each study separately. Second, the studies comprising each evaluation varied in the RE programs delivered. Therefore, a total of 31 studies were synthesized to evaluate the meta-analytic effect. The characteristics of the primary studies included in the meta-analyses are outlined below and additional information is presented in Table 1.

5.2. Study characteristics

The 31 studies included in the meta-analysis were published between 1993 and 2020.

5.2.1. Sample characteristics

The sample across the 31 independent studies totaled 25,527 participants. The size of the sample ranged from 8 to 2020 individuals across studies. As shown in Table 2, the samples comprised of couples ($k = 25$) and individuals who were in a relationship or not ($k = 6$). One study involved male participants exclusively, whereas another four studies included predominantly female participants. The remaining 26 studies included an even ratio of male to female participants. The overall mean age of participants across studies was 28.25 years, with mean age range between 16.01 and 44.7 years. The majority of the samples were comprised of participants with low socioeconomic status or income ($k = 9$), and most studies recruited samples that reported past experiences of minor to moderate couple violence ($k = 11$). However, some studies excluded participants who reported experiences of severe physical, psychological, or sexual violence in the past 6- to 12-months or across the span of the current relationship ($k = 4$).

5.2.2. Program

As shown in Table 1, relationship education programs varied across the included studies. The PREP program (and an online version of the same program [ePREP]) as well as an adaptation of PREP for individuals (Within My Reach program) were evaluated most often. Specifically, seven of the studies included in the current review evaluated the Within My Reach Program, three studies evaluated ePREP, and one study evaluated PREP. Other programs that were evaluated across multiple studies included: Loving Couples, Loving Children ($k = 7$), Love's Cradle ($k = 2$), Becoming Parents (k = 2) and the Couple CARE program ($k = 2$). The remaining relationship education programs were evaluated by one study each. Although the majority of the programs were delivered in-person, three studies delivered the ePREP program which employs a self-paced online format. Across the studies, treatment retention varied greatly from 13% to 100%, with Loving Couples, Loving Children (BSF Project), Men in Healthy Relationships, Creating Healthy Relationships and Couple CARE programs demonstrating the weakest retention across the studies.

5.2.3. Study design

As shown in Table 1, 24 of the studies included in the review were Randomized Control Trials (RCTs), and six were pre-post comparisons, and one study was a non-randomized control trial. Importantly, the Within My Reach program, whilst being evaluated six times, was evaluated using a pre-post comparison design on three of these occasions, and once using a non-RCT design, representing a lower form of evidence than the RCTs that were utilized to evaluate other programs. Follow up periods for studies ranged from immediately post-intervention up to 60 months post-intervention. The most common length of follow up was six months post-intervention.

5.2.4. Outcome measures

The most commonly utilized assessments of physical and psychological aggression were the Conflict Tactic Scale (CTS) and Revised Conflict Tactic Scale (CTS2), which were utilized in 24 studies (see Table 1). The Controlling Behaviors Scale (CBS) was utilized in two studies (Antle et al., 2011; Owen et al., 2017; see Table 1), the Continuum of Conflict and Control Relationship Scale (CCC-RS) was used in one study (Carlson, Wheeler, and Adams, 2018; see Table 1), and the Abusive Behavior Inventory was used in one study (Wong and Bouchard, 2020; see Table 1). One study asked four questions related to emotional abuse and coercion that were developed for the purpose of the study (Charles et al., 2014) and another two developed seven items pertaining to the experience of intimate partner violence (IPV) for the purpose of their study (Doss et al., 2020). One study coded semi-structured interviews that focused on relationship conflict and physical aggression to derive a IPV score on a scale ranging from 0 (no violence) to 3 (serious violence; Florsheim et al., 2011).

In terms of negative aspects of relationship functioning beyond aggression, destructive conflict behaviors was the only outcome assessed, and a variety of measures were used to assess this construct. Three studies developed study-specific assessments (two self-report: Charles et al., 2014; Doss et al., 2020; one observational assessment: Bakhurst, McGuire, and Halford, 2017), two studies used the Communication Patterns Questionnaire (CPQ, Antle et al., 2013; Braithwaite and Fincham, 2011), one study used the Reduced Sound House Questionnaire (RSHQ; Cleary Bradley, Friend, and Gottman, 2011), one study used the Continuum of Conflict and Control Relationship Scale (CCC-RS; Carlson et al., 2018).

5.3. Study quality and risk of bias

Studies generally demonstrated moderate to high study quality as assessed across the three domains of the CQC (see Table 1). Forty percent of studies fulfilled at least 66% of the criteria to achieve a moderate to high rating on the correlate score domain. All 31 studies achieved a maximum score of 3 on the risk factor score domain. Finally, 77% of studies scored a maximum score of 7 for the causal risk factor score domain. In terms of risk of bias, we evaluated bias in the seven non-RCT studies to assess the appropriateness of combining these studies as part of the overall quantitative synthesis. Six out of seven studies were found to have an overall moderate level of bias, and one study was found to have a serious level of bias (see Table 1).

5.4. Publication bias

Visual inspection of the funnel plot of studies indicated no potential publication bias with regards to relationship aggression (k = 31, Fig. 2). Confirming this, Egger's test of the intercept was not significant for the meta-analysis of relationship aggression overall (intercept = 0.88; 95% CI [−0.24, 2.20]; $p = .12$). Visual inspection of the funnel plot for destructive conflict behaviors – the only negative aspect of relationship functioning reported across studies (k = 7) – indicated no potential publication bias (Fig. 3). Egger's test of the intercept was again not significant (intercept = −1.60; 95% CI [−4.62, 1.42]; $p = .23$). Although publication bias was not detected, the overall point estimates should be interpreted with caution due to the considerable heterogeneity of the data.

Table 1
Study characteristics and study-level effect size estimates.

Study	N	Measure of relationship aggression	Study design	Intervention	Gender; Mean age (in years)	Treatment hours	Treatment sessions	Assessments	CQC correlate score	CQC risk factor score	CQC causal risk factor score	Bias	Outcome measured[a]	d[b]
Antle et al. (2011)	419 individuals	CTS-2	Pre-post	Within My Reach	Age not specified Female: 79.8% Male: 20.2% Age not specified	15	15	Baseline, post; 6-months post	4	3	3	M	Relationship aggression (overall)	0.20
Antle et al. (2013)	202 individuals	CTS-2	Pre-post	Within My Reach.	Female: 76.7% Male: 23.2% Age not specified	15	15	Baseline, post; 6-months post	3	3	3	M	Physical aggression Conflict	0.290 .25
Bakhurst et al. (2017)	32 couples	CTS short version:	RCT	Couple CARE in Uniform.	Female: 50%; 34.3 Male: 50%; 32.8	12	6	Baseline, post, 6-months post	2	3	7	–	Physical aggression Conflict	0.12– 0.69
Braithwaite and Fincham (2011)	77 couples	CTS-2	RCT	ePREP	Female: 50% Male: 50% Age not specified	6	Self-paced, online delivery	Baseline, post	3	3	7	–	Relationship aggression (overall)	0.44
Braithwaite and Fincham (2014)	52 couples	CTS-2	RCT	ePREP.	Female: 50% Male: 50% Age (overall sample only): 32.36	Duration: 6 h over 6 weeks	Self-paced, online delivery	Baseline, post, 12-months post	3	3	7	–	Relationship aggression (overall)	0.05
Carlson et al. (2018)	340 individuals	IJS and CCC-RS	Pre-post	Within My Reach.	Female: 83.9% Male: 16.1% Age (overall sample only): 41.4	Duration: 1 session a week, 4 weeks	4	Baseline, post	2	3	3	M	Controlling behavior Conflict	0.210 .24

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Table 1 (*continued*)

Study	N	Measure of relationship aggression	Study design	Intervention	Gender; Mean age (in years)	Treatment hours	Treatment sessions	Assessments	CQC correlate score	CQC risk factor score	CQC causal risk factor score	Bias	Outcome measured[a]	d[b]
Charles et al. (2014)	188 individuals	Study derived measure	Pre-post	Strong Couples-Strong Children	Female: 49.4% Male: 50.6% Age (overall sample only): 27.01	44	22	Baseline 1, Baseline 2, post, 6-months post	0	3	3	M	Psychological aggression Conflict	0.290 .29
Cleary Bradley et al. (2011)	115 couples	CTS-2	RCT	Creating Healthy Relationships	Female: 50%; 34 Male: 50%; 35	44	22	Baseline, post (0–6 months)	0	3	7	–	Conflict	0.65
Doss et al. (2020)	752 couples	Study derived measure	RCT	ePREP	Female: 52.5% Male: 47.5% Age (overall sample only): 33.19	6 online +6–12 h homework	Self-paced, online delivery	Baseline, mid-treatment, post, 2-months post, 4-months post	5	3	7	–	Physical aggression Conflict	0.240 .52
Florsheim, McArthur, Hudak, Heavin, and Burrow-Sanchez (2011)	105 couples	Interview screening	RCT	Young Parenthood Program	Female: 50%; 16.1 Male: 50%; 18.3	Not provided	8–12	Pre, post (2–3 months & 18-months post birth)	3	3	7	–	Physical aggression	0.23
Heyman et al. (2019)	368 couples	CTS-2	RCT	Couple CARE	Female: 50%; 26.76 Male: 50%; 29.31	8	8	Baseline, post, 15-months post, 24-months post.	2	3	7	–	Relationship aggression (overall)	0.07
Markman, Renick, Floyd, Stanley, and Clements (1993)	114 couples	CTS	RCT	PREP	Female: 50%; 23 Male: 50%; 24	15	5	3 years post, 4 years post, 5 years post.	2	3	7	–	Physical aggression Conflict	0.360 .11
Owen et al. (2017)	154 individuals	CTS and CBS.	Non-Random Control Trial	Within My Reach.	Female: 83.9% Male: 14.8% Age (overall sample only): 37.27	16	15	Baseline, post, 6-months post.	2	3	4	M	Relationship aggression (overall)	.30[3]

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Table 1 (*continued*)

Study	N	Measure of relationship aggression	Study design	Intervention	Gender; Mean age (in years)	Treatment hours	Treatment sessions	Assessments	CQC correlate score	CQC risk factor score	CQC causal risk factor score	Bias	Outcome measured[a]	d[b]
Rey-Anacona et al. (2014)	4 couples	Semi-structured interview	Pre-post	Dating violence education program	Female: 50% Male: 50% Age (overall sample only): 20	20	10	Baseline, post	2	3	3	S	Relationship aggression (overall)	1.12
SHM | Bronx	683 couples	CTS-2 and PMWI	RCT	Loving Couples, Loving Children	Female: 50% Male: 50% Age (overall sample only): 35.6	24	11	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	0.002
SHM | Oklahoma City	1001 couples	CTS-2 and PMWI	RCT	Becoming Parents	Female: 50% Male: 50% Age (overall sample only): 27.5	30	16	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	0.004
SHM | Orlando	709 couples	CTS-2 and PMWI	RCT	For our future, for our family	Female: 50% Male: 50% Age (overall sample only): 31.5	30	18	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	−0.03
SHM | Pennsylvania	567 couples	CTS-2 and PMWI	RCT	Within Our Reach	Female: 50% Male: 50% Age (overall sample only): 33.2	28	22–30	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	0.01
SHM | Seattle	554 couples	CTS-2 and PMWI	RCT	Becoming Parents	Female: 50% Male: 50% Age (overall sample only): 27.0	30	15	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	−0.06
SHM | Shoreline	679 couples	CTS-2 and PMWI	RCT	Loving Couples, Loving Children	Female: 50% Male: 50% Age (overall sample only): 32.4	24	24	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	0.02

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Table 1 (*continued*)

Study	N	Measure of relationship aggression	Study design	Intervention	Gender; Mean age (in years)	Treatment hours	Treatment sessions	Assessments	CQC correlate score	CQC risk factor score	CQC causal risk factor score	Bias	Outcome measured[a]	d[b]
SHM | Texas	691 couples	CTS-2 and PMWI	RCT	Within Our Reach	Female: 50% Male: 50% Age (overall sample only): 33.5	28	15	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	0.17
SHM | Witchita	670 couples	CTS-2 and PMWI	RCT	Within Our Reach	Female: 50% Male: 50% Age (overall sample only): 31.20	28	12	Baseline, 12- and 30-months post	2	3	7	–	Relationship aggression (overall)	0.03
Wong and Bouchard (2020)	46 individuals	ABI	Pre-post	Men in Healthy Relationships Program	Male: 100%; 44.7	50	20	Baseline, post, 6-months post	0	3	4	M	Relationship aggression (overall)	0.68
Wood, Moore, Clarkwest, Killewald, and Monahan (2012) – BSF | Atlanta	903 couples	CTS-2 (physical assault subscale)	RCT	Loving Couples, Loving Children	Female: 50%; 23 Male: 50%; 25	42	Not reported	15-months post recruitment	3	3	7	–	Physical aggression	0.02
Wood et al. (2012) – BSF | Baltimore	602 couples	CTS-2 (physical assault subscale)	RCT	Loving Couples, Loving Children	Female: 50%; 23 Male: 50%; 26	42	Not reported	15-months post recruitment	3	3	7	–	Physical aggression	−0.09
Wood et al. (2012) – BSF | Baton Rouge	652 couples	CTS-2 (physical assault subscale)	RCT	Loving Couples, Loving Children	Female: 50%; 22 Male: 50%; 25	42	Not reported	15-months post recruitment	3	3	7	–	Physical aggression	0.13
Wood et al. (2012) – BSF | Florida	695 couples	CTS-2 (physical assault subscale)	RCT	Loving Couples, Loving Children	Female: 50%; 22 Male: 50%; 25	42	Not reported	15-months post recruitment	3	3	7	–	Physical aggression	−0.02
Wood et al. (2012) – BSF | Houston	405 couples	CTS-2 (physical assault subscale)	RCT	Love's Cradle	Female: 50%; 25 Male: 50%; 27	42	Not reported	15-months post recruitment	3	3	7	–	Physical aggression	0.09
Wood et al. (2012) – BSF | Indiana	466 couples	CTS-2 (physical assault subscale)	RCT	Loving Couples, Loving Children	Female: 50%; 23 Male: 50%; 26	42	Not reported	15-months post recruitment	3	3	7	–	Physical aggression	−0.02
Wood et al. (2012) – BSF | Oklahoma City	1010 couples	CTS-2 (physical assault subscale)	RCT	Becoming Parents	Female: 50%; 23 Male: 50%; 25	30	Not reported	15-months post recruitment	3	3	7	–	Physical aggression	0.03

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Table 1 (*continued*)

Study	N	Measure of relationship aggression	Study design	Intervention	Gender; Mean age (in years)	Treatment hours	Treatment sessions	Assessments	CQC correlate score	CQC risk factor score	CQC causal risk factor score	Bias	Outcome measured[a]	d[b]
Wood et al. (2012) – BSF | San Angelo	342 couples	CTS-2 (physical assault subscale)	RCT	Love's Cradle	Female: 50%; 22 Male: 50%; 24	42	Not reported	15-months post recruitment	2	3	7	–	Physical aggression	0.01

RCT = Randomized Control Trial; CQC = Cambridge Quality Checklist; M = Moderate; S = SeriousABI = Abusive Behavior Inventory; CTS = Conflict Tactics Scale; CTS-2 = Conflict Tactics Scale – 2; PMWI = Psychological Maltreatment of Women Inventory.

[a] For studies that assessed multiple aspects of relationship aggression or positive relationship functioning, a pooled effect size was estimated. Estimation of relationship aggression overall entailed pooling of any two or more aspects of relationship aggression (e.g., physical, psychological, controlling behavior). Positive relationship functioning overall entailed pooling effects across any two or more aspects of constructive/adaptive relationship processes and qualities (e.g., relationship satisfaction, relationship skills).

[b] All estimates of Cohen's d are coded so that positive values reflect positive outcomes (e.g., reductions in relationship aggression and conflict, increases in positive relationship functioning).

5.5. Sensitivity analyses

The one-study-removed analyses indicated there were no outlier studies included in the meta-analyses for relationship aggression overall and conflict behavior. Therefore, sensitivity analysis revealed that the average effect sizes estimated in these meta-analyses were robust and did not depend on any one study alone (see Appendix, Figs. A3 to A4).

5.6. Synthesis of results

5.6.1. Relationship aggression: Physical aggression, psychological abuse, and controlling behavior

All but one study (which focused on destructive conflict, but assessed baseline relationship aggression) measured the effect of relationship education on reducing various forms of relationship aggression, including, physical aggression, psychological abuse, and/or controlling behavior. When these outcomes are combined into an overall relationship aggression outcome, relationship education programs were found to demonstrate a small but significant reduction in relationship aggression (d = 0.11, p = .001 [k = 30]) (see Table 2; see forest plot Appendix A Fig. A1). We also evaluated the meta-analytic effect of relationship education programs on each type of relationship aggression. Relationship education programs significantly reduced physical aggression (d = 0.09, p = .01 [k = 27]) and controlling behavior (d = 0.22, p < .0001 [k = 3]), but had no significant effect on reducing psychological abuse (d = 0.001, p = .98 [k = 14]) (see Table 2).

5.6.2. Negative aspects of relationship functioning (conflict behavior)

The only aspect of negative relationship functioning that was assessed across studies was conflict behavior. Relationship education programs demonstrated a significant reduction in conflict behavior (d = 0.31, p < .0001 [k = 7]) (see Table 2; see forest plot Appendix A Fig. A2.)

5.7. Subgroup analyses: Methodological characteristics

Where subgrouping allowed (i.e., $k > 2$), subgroup analyses were conducted to determine whether various methodological characteristics, specifically, sample characteristics, program characteristics, or study characteristics moderated the effect of relationship education programs on each outcome. However, it is important to note that all subgroup analysis that were conducted and reported we subjected to power analysis which took into account the number of studies, sample size, effect size and heterogeneity (e.g., Griffn, 2021). Thus, all reported subgroup analysis were estimated to have a power ≥ 0.80. As outlined in the subsections below, subgroup analyses revealed several significant effects for various methodological characteristics on the estimated effect sizes (see Table 3). Z-tests were conducted to determine whether differences in subgroup estimated effects were significant (see Appendix B).

5.7.1. Sample characteristics

Differences in baseline aggression. Subgroup analyses revealed differences in participants' reported level of aggression at baseline significantly moderated the effect of RE on relationship aggression. Specifically, participants that reported moderate-to-severe levels of aggression at baseline experienced significantly greater reductions for relationship aggression overall (d = 0.57, p = .001 [k = 4]) compared to participants that reported none-to-low levels of aggression (d = 0.08, p = .004 [k = 27]; z = 8.15, p = .004) (see Table 3 and Appendix B). When examining specific aspects of relationship aggression, participants who reported moderate-to-severe levels of aggression at baseline reported significantly greater reductions in physical aggression (d = 0.66, p = .01 [k = 4]) and psychological abuse (d = 0.85, p < .001 [k = 3]) compared to those who reported none-to-low aggression (physical aggression: d = 0.07, p = .009 [k = 23], z = 11.09, p = .001; psychological abuse: d = −0.04, p = .17 [k = 12], z = 23.60, p < .001; see Table 3 and Appendix

Table 2
Random effects results of meta-analyses.

Outcome	Number of studies	d[a]	95% CI	σ^2	p	Heterogeneity	
						Q-statistic (I^2)	p
Relationship aggression (overall)[a]	30	0.11	0.05–0.17	0.01	0.0008	49.49 (43.41)	0.007
Physical aggression	27	0.09	0.02–0.16	0.01	0.01	40.86 (38.82)	0.02
Psychological abuse	14	0.001	−0.07–0.07	0.009	0.98	56.85 (77.13)	<0.0001
Controlling behavior	3	0.22	0.11–0.33	0.00	0.0001	1.98 (0.00)	0.37
Negative relationship functioning (destructive conflict)	7	0.31	0.16–0.46	0.02	0.0001	15.90 (62.26)	0.01

[a] Pooled estimates for BSF and SHM Oklahoma City programs as these studies report on the same sample.

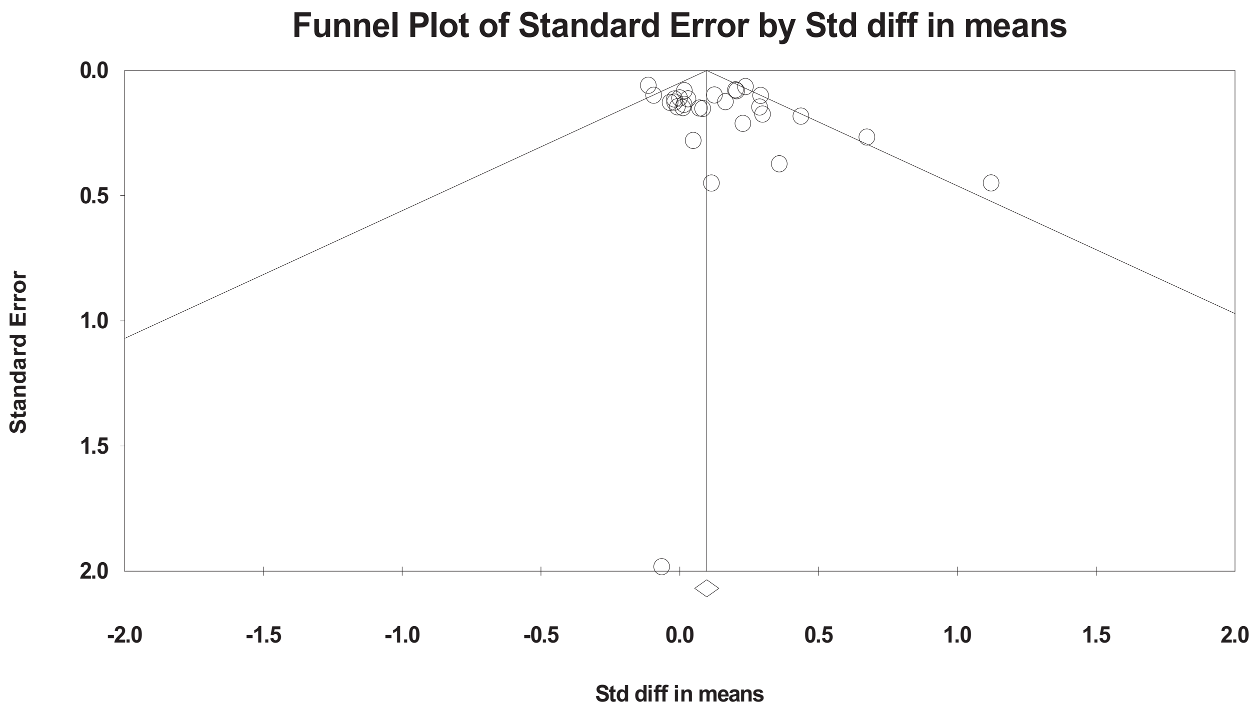

Fig. 2. Publication bias for relationship aggression overall.

B). No subgroup analysis could be conducted on controlling behavior. In terms of negative aspects of relationship functioning (conflict behaviors only) no significant differences were found between subgroups for reductions in conflict (see Table 3 and Appendix B).

Gender. Eleven studies reported effects separately for subsamples of males and females, allowing for a subgroup analysis of gender. Although the magnitude of effects of relationship education on reducing relationship aggression was larger for men ($d = 0.13$, $p = .03$ [$k = 11$]) than women ($d = 0.05$, $p = .44$ [$k = 11$]) (see Table 3), Z-tests revealed that the difference in effect sizes was not significant ($z = 0.89$, $p = .37$) (see Appendix B). This finding regarding gender was consistent across outcomes of physical aggression and psychological abuse (no subgroup analyses could be conducted for controlling behavior). No subgroup analyses for gender could be conducted on conflict behavior (see Appendix B).

5.7.2. *Program characteristics*

Method of program delivery. The estimated effect of online program delivery was statistically significant and significantly larger in magnitude than for face-to-face program delivery for reduction in relationship aggression overall (online: $d = 0.25$, $p < .001$ [$k = 3$] vs face-to-face: $d = 0.09$, $p = .006$ [$k = 27$]; $z = 2.52$, $p = .02$) as well as for psychological aggression (online: $d = 0.45$, $p = .005$ [$k = 3$] vs face-to-face: $d = -0.02$, $p = .48$ [$k = 12$]; $z = 2.91$, $p = .004$). Online delivery was associated with statistically significant reductions in physical aggression ($d = 0.22$, $p = .04$ [$k = 3$]), but not for face-to-face delivery ($d = 0.07$, $p = .44$ [$k = 24$]). However, the magnitude of these associations did not differ for method of delivery ($z = 1.30$, $p = .19$). There were not enough studies to conduct subgroup analyses of program delivery method on conflict behavior.

Dose-response relationship. In order to determine whether the amount of time spent in relationship education programs moderated outcomes, we created subgroups where possible. Subgroup analyses suggest the amount of time spent in a relationship education program is associated with the effect of the program on relationship aggression overall. Specifically, programs 6–12 h in duration were associated with significant reductions in relationship aggression ($d = 0.22$, $p < .001$ [$k = 6$]) as were programs of 13–24 h in duration s ($d = 0.27$, $p < .001$ [$k = 6$]). Programs of greater duration were not found to be associated with significant reductions in regression overall (see Table 3). When examining different aspects of aggression, programs were more effective at reducing physical and psychological relationship aggression when 24 h or less in duration (physical aggression: 6–12 h: $d = 0.21$, $p < .001$ [$k = 5$]; 13–24 h: $d = 0.28$, $p = .04$ [$k = 4$]; psychological aggression: 6–12 h: $d = 0.27$, $p = .07$ -marginal significance - [$k = 3$]; 13–24 h: $d = 0.30$, $p = .006$ [$k = 3$]) (see Table 3 and Appendix B). In contrast, programs that were >25 h in duration were associated with small and non-significant decreases in relationship aggression overall ($d = 0.001$, $p = .99$ [$k = 16$]) and physical aggression ($d = 0.03$, $p = .43$ [$k = 16$]), but small increases in psychological aggression ($d = -0.08$, $p < .0001$ [$k = 8$]). Differences

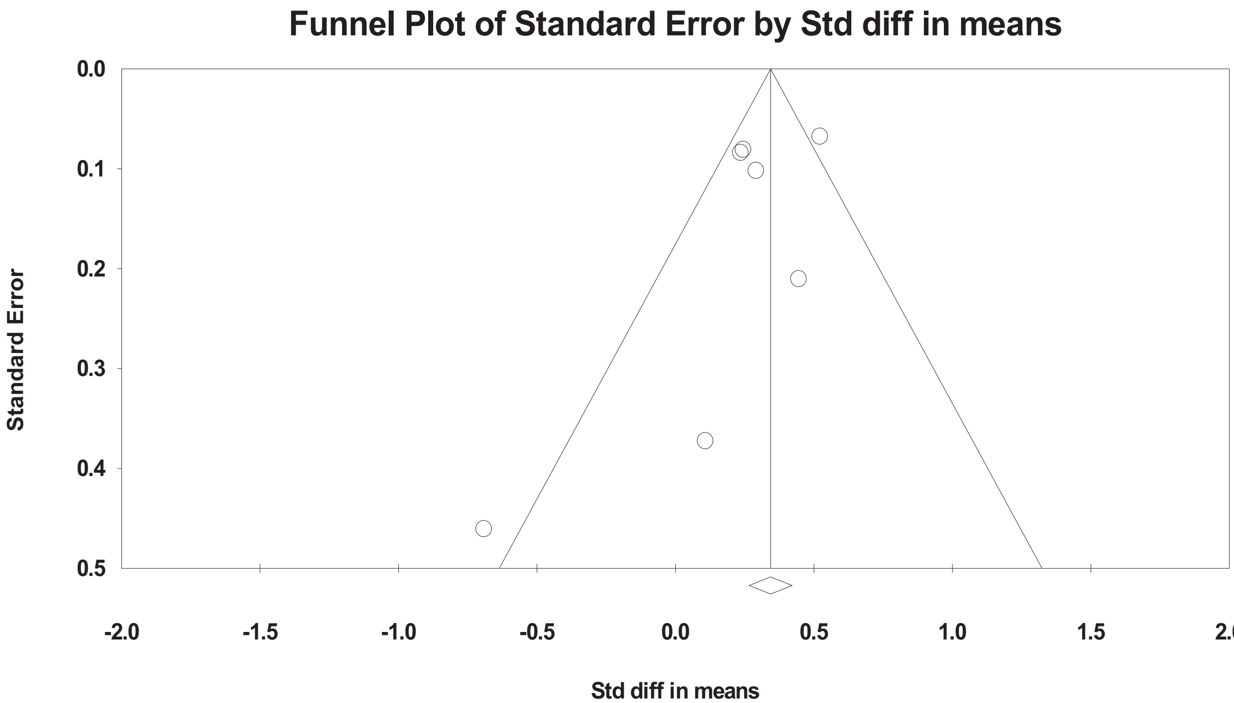

Fig. 3. Publication bias for negative relationship functioning – destructive conflict behavior.

in magnitude of effect sizes for program dosage on relationship aggression overall were found between found between programs of 6–12 h duration when compared to those over 25 h ($z = 4.10$, $p < .001$) and between programs of 13–24 h duration and those over 25 h ($z = 4.31$, $p < .001$) (see Appendix B). For physical aggression significant differences were found between programs of 6–12 h duration and those over 25 h ($z = 2.68$, $p = .007$). For psychological abuse, differences were found between programs of 6–12 h duration when compared to those over 25 h ($z = 2.32$, $p = .02$) and between programs of 13–24 h duration and those over 25 h ($z = 3.42$, $p = .001$) (see Appendix B). Again, there were not enough studies to conduct subgroup analyses for dose-response on conflict behavior.

5.7.3. Program

The type of relationship education programs implemented across studies varied. Therefore, where possible, subgroup analyses were conducted to determine whether the type of program moderated the estimated effect for relationship aggression (there were not enough studies to conduct such analyses for conflict behavior). The programs for which subgroup analysis could be conducted were PREP and ePREP (including Within my Reach and Within Our Reach), Couple CARE, Loving Couples, Loving Children, Love's Cradle, and Becoming Parents. PREP and ePREP ($d = 0.20$, $p < .001$, [$k = 11$]) associated with significant reductions in overall relationship aggression. Couple CARE ($d = 0.08$, $p = .59$ [$k = 2$]), Loving Couples, Loving Children ($d = 0.01$, $p = .83$ [$k = 7$]), Love's Cradle ($d = 0.04$, $p = .73$ [$k = 2$]) and Becoming Parents ($d = -0.11$, $p = .07$ [$k = 2$]) were not found to be associated with significant reductions in relationship aggression. However, no significant differences were found in the effect size estimates between the type of relationship education program delivered and relationship aggression overall, or the specific outcomes of physical and psychological aggression (see Table 3 and Appendix B). Finally, subgroup analysis of PREP and ePREP revealed a reduction in conflict behavior ($d = 0.33$, $p < .001$ [$k = 4$]).

5.8. Study design characteristics

5.8.1. Follow-up period

Subgroup analyses revealed that assessment of outcomes at different periods post program completion (i.e., immediately following program completion, 1–6 months follow-up, 7–12 months follow-up, 13–24 months follow-up, and 30+ months after program completion) moderated the effect size of relationship aggression overall (see Appendix B). Specifically, the effect of RE programs was significantly larger immediately post program ($d = 0.30$, $p < .001$ [$k = 6$]) and 1–6 months follow-up ($d = 0.2330$, $p < .001$ [$k = 7$]) compared to the effect size at 13–24 months follow-up ($d = 0.02$, $p = .56$ [$k = 9$]) and 30+ months follow up ($d = -0.03$, $p = .51$ [$k = 9$]) ($zs = 2.98$ to 4.45, $ps = 0.003$ to <0.001; see Table 3 and Appendix B). In terms of physical aggression, the effect of RE programs did not differ between immediately post program completion ($d = 0.33$, $p = .03$ [$k = 5$]) and at 1–6 months follow-up ($d = 0.21$, $p < .001$ [$k = 5$]) ($z = 0.76$, $p = .45$). However, the effect size at 1–6 months follow-up was significantly larger than at 13–24 months follow-up ($d = 0.02$, $p = .60$ [$k = 9$]) ($z = 2.97$, $p = .003$) and at 30+ months follow-up ($d = 0.04$, $p = .15$ [$k = 9$]) ($z = 2.27$, $p = .02$). In relation to psychological aggression, there were no differences in effect sizes between immediately post program completion ($d = 0.59$, $p = .006$ [$k = 3$]) and 1–6 months follow-up ($d = 0.28$, $p = .002$ [$k = 3$]) ($z = 1.33$, $p = .19$). However, significant differences in the effect sizes were found when comparing immediate post program completion and 1–6 months follow-up with 7–12 ($d = -0.06$, $p = .03$) and 30+ months follow-up periods ($d = -0.09$, $p < .001$) ($zs = 3.04$–3.91, $ps = 0.002$ - < 0.001 (see Table 3 and Appendix B).

5.8.2. Study design

Studies were grouped by study design (e.g., pre-post design or randomized control trials). Subgroup analyses revealed pre-post study designs resulted in significantly greater reductions in relationship aggression overall ($d = 0.27$, $p < .0001$ [$k = 6$]) and psychological aggression ($d = 0.35$, $p < .001$ [$k = 4$]) than those reported in randomized control trials (aggression overall: $d = 0.05$, $p = .10$ [$k = 23$], $z = 4.24$, $p < .0001$; psychological aggression: $d = -0.08$, $p < .001$ [$k =$

Table 3
Subgroup analyses.

Subgroup category	Outcome	Subgroup level	*k*	*d*	(95% CI)	σ^2	*p*	Heterogeneity Q-statistic (I^2)	Heterogeneity *p*
Baseline level of aggression	Relationship aggression (overall)	None-to-low	26	0.08	0.03–0.14	0.001	0.04	5.67 (47.08)	0.13
		Moderate-to-severe	4	0.57	0.24–0.90	0.03	0.001	34.00 (23.54)	0.14
	Physical aggression	None-to-low	23	0.07	0.01–0.13	0.001	0.02	27.14 (18.94)	0.21
		Moderate-to-severe	3	0.66	0.32–1.01	0.03	<0.0001	2.07 (3.31)	0.36
	Psychological abuse	None-to-low	12	−0.04	−0.09–0.02	0.001	0.17	34.23 (67.86)	0.0003
		Moderate-to-severe	3	0.85	0.50–1.20	0.03	<0.0001	1.12 (0.00)	0.57
	Controlling Behavior	None-to-low	3	0.20	0.05–0.36	0.006	0.01	2.73 (26.78)	0.26
		Moderate-to-severe	1	N/A					
	Negative relationship functioning (conflict)	None-to-low	4	0.30	0.11–0.49	0.009	0.002	13.46 (77.71)	0.004
		Moderate-to-severe	2	0.40	0.20–0.59	0.01	0.0001	0.08 (0.00)	0.78
Gender	Relationship aggression (overall)	Men	9	0.13	0.01–0.26	0.003	0.03	9.65 (17.11)	0.29
		Women	10	0.05	−0.08–0.18	0.004	0.44	17.38 (48.21)	0.04
	Physical aggression	Men	9	0.14	0.01–0.24	0.003	0.04	10.58 (24.37)	0.23
		Women	10	0.03	−0.09–0.15	0.004	0.64	12.70 (29.11)	0.18
	Negative relationship functioning (conflict)	Male	2	−0.09	−1.50 - 1.33	0.52	0.90	7.98 (87.46)	0.005
		Female	2	0.002	−78 - 0.79	0.16	1.0	2.72 (63.18)	0.10
Method of program delivery	Relationship aggression (overall)	Online	3	0.25	0.13–0.37	0.00	<0.0001	1.56 (0.00)	0.46
		Face-to-Face	26	0.09	0.03–0.15	0.006	0.03	40.61 (38.45)	0.03
	Physical aggression	Online	3	0.22	0.01–0.44	0.01	0.04	2.63 (24.13)	0.27
		Face-to-Face	12	0.07	0.002–0.14	0.04	0.04	31.20 (29.48)	0.09
	Psychological abuse	Online	2	0.45	0.14–0.76	0.16	0.01	0.67 (0.00)	0.41
		Face-to-Face	13	−0.02	−0.09–0.04	0.03	0.48	45.81 (75.99)	<. 0001
	Controlling Behavior	Online	0	*NA*					
		Face-to-Face	3	0.22	0.11–0.33	0.003	0.0001	1.98 (0.00)	0.37
	Conflict	Online	1	*NA*					
		Face-to-Face	6	0.25	0.15–0.35	0.003	<0.0001	5.37 (6.92)	0.37
Dose-response relationship	Relationship aggression (overall)	6–12 h	6	0.22	0.13–0.31	0.002	<0.0001	2.83 (0.00)	0.73
		13–24 h	6	0.27	0.16–0.37	0.003	<0.0001	4.41 (0.00)	0.49
		25–50 h	16	0.001	−0.06–0.06	0.0008	0.99	14.17 (0.00)	0.51
	Physical aggression	6–12 h	5	0.21	0.10–0.33	0.03	0.004	3.62 (0.00)	0.46
		13–24 h	4	0.28	0.02–0.53	0.02	0.04	4.87 (38.36)	0.18
		25–50 h	16	0.03	−0.04–0.10	0.001	0.43	18.06 (16.93)	0.26
	Psychological abuse	6–12 h	3	0.27	−0.02–0.57	0.02	0.07	3.61 (44.69)	0.16
		13–24 h	3	0.30	0.09–0.51	0.11	0.006	4.06 (50.73)	0.13
		25–50 h	8	−0.08	−0.12 – -0.05	0.0004	<0.0001	12.04 (41.85)	0.10
	Negative relationship functioning (conflict)	6–12 h	3	0.27	−0.07–0.60	0.03	0.12	12.60 (84.13)	0.002
		13–24 h	3	0.26	0.14–0.38	0.004	<0.0001	0.29 (0.00)	0.86
		25–50 h	1	*NA*					

(*continued on next page*)

Table 3 (*continued*)

Subgroup category	Outcome	Subgroup level	*k*	*d*	(95% CI)	σ^2	*p*	Heterogeneity	
								Q-statistic (I^2)	*p*
Program	Relationship aggression (overall)	Prep and ePrep	11	0.20	0.13–0.27	0.001	<0.0001	6.82 (0.00)	0.74
		Loving Couples, Loving Children	7	0.01	−0.07–0.09	0.002	0.83	2.45 (0.00)	0.88
		Becoming Parents	2	−0.11	−0.23–0.01	0.008	0.92	0.00 (0.00)	0.93
		CoupleCare	2	0.08	−0.21–0.36	0.02	0.59	0.01 (0.00)	0.93
		Love's Cradle	2	0.04	−0.17–0.25	0.01	0.73	0.18 (0.00)	0.67
	Physical aggression	Prep and ePrep	9	0.21	0.13–0.29	0.002	< 0.0001	4.74 (0.00)	0.79
		Loving Couples, Loving Children	7	0.02	−0.06–0.10	0.002	0.63	2.83 (0.00)	0.83
		Becoming Parents	2	−0.11	−0.23–0.01	0.004	0.08	0.001 (0.00)	0.98
		CoupleCare	2	0.07	−0.24–0.37	0.03	0.67	0.01(0.00)	0.91
		Love's Cradle	2	0.04	−0.17–0.25	0.01	0.73	0.18 (0.00)	0.67
	Psychological abuse	Prep and ePrep	6	0.01	−0.11– -0.13	0.06	0.84	27.49 (81.81)	<0.0001
		Loving Couples, Loving Children	2	−0.08	−0.13 – -0.03	0.001	0.002	0.49 (0.00)	<0.0001
	Controlling Behavior	Prep and ePrep	3	0.22	0.11–0.33	0.06	0.0001	1.98 (0.00)	0.37
	Negative relationship functioning (conflict)	Prep and ePrep	4	0.33	0.15–0.51	0.009	0.0004	10.28 (70.81)	0.02
	Relationship aggression (overall)	Post program	6	0.31	0.16–0.38	0.003	0.004	6.84 (26.92)	0.24
		1–6 months post	7	0.230	0.15–0.31−	0.0070	<0.00010	1.97 (0.00)	0.920
		7–12 months post	9	.05	0.04–0.13	.002	.28	12.45 (35.74)	.13
		13–24 months post	9	0.02−	−0.06–0.10−	0.0020	0.560	3.45 (0.00)	0.900
		30+ months post	9	0.03	0.11–0.05	.002	.51	6.52 (0.00)	.59
	Physical aggression	Post program	5	0.33	0.03–0.64	0.15	0.03	9.90 (59.60)	0.04
		1–6 months post	5	0.210	0.11–0.30−	0.0020	<0.00010	1.38 (0.00)	0.850
		7–12 months post	7	.11	0.002–0.21	.06	.06	3.65 (0.00)	.72
Follow up period		13–24 months post	9	0.020	−0.06–0.10−	0.040	0.180	3.39 (0.00)	0.910
		30+ months post	9	.04	0.08–0.15	.004	.55	10.12 (20.94)	.26
	Psychological abuse	Post program	3	0.59	0.17–1.01	0.21	0.05	2.71 (26.11)	0.26
		1–6 months post	3	0.28−	0.10–46−	0.010	0.0020	2.39 (16.43)	0.300
		7–12 months post	9	0.06	0.12–0.006	.03	.03	30.90 (74.11)	.0001
		13–24 months post	1	*NA*−					
		30+ months post	7	0.09	−0.11 – -0.06	0.0002	<0.0001	2.94 (0.00)	0.82
	Negative relationship functioning (conflict)	Post program	3	0.39	−0.04–0.59	0.03	0.09	11.44 (82.51)	0.003
		1–6 months post	3	0.40	0.17–0.63	0.01	0.0005	7.46 (73.19)	0.02
	Relationship aggression (overall)	RCT	22	0.05	−0.01–0.10	0.001	0.10	25.42 (0.00)	0.23
		Pre-post	6	0.28	0.16–0.40	0.06	<0.0001	7.10 (29.53)	0.21
	Physical aggression	Pre-post	3	0.58	0.03–1.12	0.08	0.04	7.31 (72.65)	0.03
		RCT	22	0.06	−0.004–0.12	0.001	0.21	26.02 (50.38)	0.11
Study design	Psychological abuse	Pre-post	4	0.35	0.14–0.56	0.01	0.02	6.05 (50.38)	0.11
		RCT	3	−0.08	−0.11 – -0.03	0.001	0.002	17.84 (49.54)	0.04
	Controlling Behavior	Pre-post	2	0.19	0.08–0.31	0.003	0.001	0.06 (0.00)	0.81
		RCT	0	*NA*					
	Negative relationship functioning (conflict)	Pre-post	3	0.26	0.16–0.36	0.003	<0.0001	0.19 (0.00)	0.91
		RCT	4	0.29	−0.08–0.66	0.04	0.13	7.86 (61.82)	0.05

11], z = 3.95, $p < .001$) (see Table 3, Appendix B). However, there was no significant difference between magnitude of effects reported in pre-post designs and randomized control trials for physical aggression and conflict behavior (see Table 3, Appendix B). Given the significant difference in the effect size of RE on relationship aggression by study design, we conducted an additional analysis that was more specific in nature. In this analysis we compared only the effect size of the control groups of RCT studies with the effect size of pre-post studies. The reason for this analysis was to rule out whether the larger effect size evidenced in pre-post studies reflected spontaneous recovery in relationship aggression rather than an effect of the intervention. If indeed the control group studies of RCTs demonstrated little reduction in relationship aggression over time, then one can be more confident that the interventions in pre-post study designs yield some shift in relationship aggression. Given that all pre-post studies reported mean changes in relationship aggression, we limited the subgroup analysis to include the control groups of RCTs that also reported mean differences (k = 11). The subgroup analysis revealed that RE pre-post studies resulted in significantly larger reductions in relationship aggression overall (d = 0.27, $p <$ .001 [k = 6]) compared to the control groups of RCTs (d = 0.06, $p = .59$; [k = 5]; z = 2.59, $p = .01$).

6. Discussion

The primary aim of this meta-analysis was to determine whether relationship education programs delivered to individuals or couples are effective at reducing relationship aggression. Importantly, the meta-analysis focused on determining the role of various methodological characteristics to further understand the effects of RE on relationship aggression. The secondary aim was to determine the extent to which RE programs reduce negative aspects of relationship functioning, especially amongst those individuals and couples that report moderate to high levels of relationship aggression. The aims of the current meta-analysis extend on past reviews in a number of important ways. First, the review evaluated RE program effectiveness on different types of aggression. Second, it examined whether RE program effectiveness differs by level of aggression on program entry. Third it examined the role of a number of methodological characteristics which have not been examined before (e.g., program, program dosage, sample characteristics). Finally, it examined whether RE programs affect negative aspects of relationship functioning for those who vary in relationship aggression on program entry. These extensions provide important insights into whether there are particular types of relationship aggression, (e.g., physical aggression, psychological aggression) that are more amenable to change than others. Furthermore, these extensions guide an understanding of subpopulation characteristics (e.g., existing levels of aggression, gender) that can make some individuals and couples more likely to experience gains in terms of reductions in relationship aggression from participating in RE programs. Our examination of the effects of study and program design features can inform a deeper understanding as to the effectiveness of RE for relationship aggression.

Our meta-analysis included a total of 31 studies that were rated as moderate to high study quality. The findings demonstrated that overall, RE is significantly associated with small reductions in relationship aggression. Specifically, participating in RE reduces both reported physical abuse, controlling behaviors, but psychological abuse. However, participants who reported moderate to severe relationship aggression upon entering the RE program demonstrated large reductions in physical and psychological aggression – reductions that were between nine to twenty times the magnitude compared to those who reported minimal to low relationship aggression upon program entry. Thus, it appears that RE program can yield large gains for those who experience moderate to severe relationship aggression. Furthermore, in terms of the negative aspects of relationship functioning, RE is associated with large and significant decreases in negative relationship functioning – namely – conflict behavior.

The findings suggest that participants who report moderate to severe relationship aggression upon program entry have more room to experience reductions in relationship aggression. These findings are important as they are in direct contrast with suggestions that participants reporting moderate to severe relationship aggression upon program entry may be the most resistant, or experience the least gains, because the level of aggression may be more difficult to manage and may be more entrenched within one's relationship (e.g., Bradford et al., 2011).

As noted by Hawkins and Erickson (2015), many couples that enter RE programs report relationship distress, but it is these couples who may be more motivated to use the lessons learned in these programs to enhance their relationship. For those who enter RE programs, and have experienced relationship aggression, and/or are from social disadvantaged backgrounds, RE programs may be the only option to receive support to address relationship issues, as the financial cost or lack of accessibility is likely to make attending relationship therapy prohibitive. Though RE may well have important benefits for such people, RE is not therapeutic in nature, rather, it is preventative education.

On this note, it is important to highlight that there appears to be an increased interest amongst some advocates of RE to move towards a more clinical model. The reason for this shift is to address the increasing number of moderately and highly distressed couples seeking help in the form of RE (Bradford, Hawkins, and Acker, 2015), a number of whom experience relationship aggression. However, there are concerns that such shifts may inadvertently blur the distinction between RE and clinically-oriented approaches to relationship distress such as couples therapy (Markman and Ritchie, 2015). As Markman and Ritchie (2015) note, "*Part of this blurring occurs because many leaders in the [Couple] RE field and many [C]RE service providers have a clinical background and often are therapists themselves. This presents a major risk of unintentionally sliding into a clinical couples therapy approach without making an intentional choice to do so.*" (p. 656). Thus, program developers and facilitators need to be clear on whether the delivery of any given RE program should be adapted to reflect a more clinical focus to more adequately address relationship distress and aggression in the long term. If there is an explicit goal to broaden any given RE program to have more of a clinical focus, then the long-term benefits of such RE programs may well be enhanced in several ways. These can include providing follow-up or booster sessions after program completion, or supplementing program completion with brief therapeutic or coaching sessions to ensure individuals and couples continue to implement the skills developed through the program. Such approaches are currently implemented in online relationship programs such as the OurRelationship Program (Doss et al., 2016, 2020). This program is therapeutic in nature offering an integration of RE components with a self-paced derivative of Integrative Behavioral Couple Therapy (Christensen et al., 2020). However, to our knowledge, the OurRelationship Program represents one of only a few online programs taking a considered approach regarding the integration of RE with couples therapy.

Subgroup analyses revealed that there were no statistical differences in effect sizes between men and women. This is consistent with the findings of past meta-analyses suggesting that men and women tend to experience similar outcomes when taking part in RE (Hawkins et al., 2008; Pinquart and Teubert, 2010). Subgroup analysis regarding the effectiveness of RE at different assessment periods post program completion suggest that the gains made from participating in RE programs are greatest immediately post program completion, with effectiveness especially attenuated for physical and psychological aggression anywhere from six to 30 months follow-up. These findings reflect a similar attenuation in effects over time found for therapeutic interventions more generally (e.g., Weisz, McCarty, and Valeri, 2006). Thus, subsequent points of intervention post-program completion may be required in the first six months after program completion and, then again, across the first year post program completion. Offering such continued support to program completers may be deemed as a large time and cost investment for RE providers and governments involved in the public funding of RE programs. However, our findings regarding dosage response suggest that programs ranging between 6 and 24 h produce significant reductions in overall relationship compared to programs that are >24 h in total.

These findings may suggest that there is a likely cost-benefit for implementing shorter programs, and that the cost savings derived from rolling out shorter programs may be re-purposed to provide follow-up booster or coaching sessions. However, to make informed decisions regarding dosage response and the possible benefits afforded by providing follow-up booster sessions and coaching calls, studies would need to be conducted that systematically vary and compare dosage responses of programs (Hawkins and Erickson, 2015) as well as the timing, frequency, and length of support provided to program completers.

Another consideration which has cost-benefit implications regarding the scaling up and provision of RE programs to address negative outcomes such as relationship aggression relates to the mode of delivery. Our findings suggest that RE programs delivered face-to-face yield small but statistically significant reductions in relationship aggression overall as well significant reductions in conflict behavior. In contrast, online delivery of RE (specifically, ePREP) produced moderate and statistically significant effect sizes for relationship aggression overall as well as physical and psychological aggression. Although it appears that online RE delivery yields reliable reductions in relationship aggression, it is important to note that the estimated effects are based three studies (with substantial variation in effect sizes). Having said this, given the flexibility offered by online e-health, and the dramatic increase in e-therapy as a function of the COVID-19 pandemic, there is a likely need and imperative to upscale the dissemination of such RE programs within the virtual setting (Doss and Hatch, 2022; Karantzas et al., 2022). It is therefore critical that continued evaluations of online RE programs are conducted to produce a larger evidence-base on which to estimate the effectiveness of these programs.

In terms of program, the largest number of studies employed PREP or an adapted version for either online delivery (ePREP) or for delivery to individuals (Within my Reach) (k = 11), followed by Loving Couples, Loving Children (k = 7). Very few other interventions included in this meta-analysis were evaluated in more than two studies. Thus, our estimation of effect sizes and subgroup analyses for RE program is limited to PREP (and its derivatives), Loving Couples, Loving Children, Love's Cradle, Becoming Parents, and Couple CARE. As part of the analyses that we were able to conduct, PREP programs (including ePREP and Within my Reach) were found to significantly reduce overall relationship aggression – including psychological aggression as well as conflict behaviors.

The effects found in this synthesis regarding RE programs, as they relate to relationship aggression, are largely limited to PREP programs, and suggest that PREP, and its adaptations, are effective at reducing abusive behaviors. This may well be because over the course of the development and adaptation of PREP, there has been an increasing emphasis and inclusion of content that addresses issues of relationship aggression. For instance, the Within My Reach Program (an adaptation of PREP delivered to individual not in current relationships) includes curriculum topics that focus on identifying and dealing with patterns of aggression as well as strategies to exit an unsafe relationship (Rhoades and Stanley, 2011). Therefore, one important consideration regarding the future application of RE programs to addressing issues of relationship aggression is that content be included that is specific to relationship aggression and draws participants' awareness to the signs of aggression, but also, specific knowledge and strategies on how to effectively and safely manage or exit relationships characterized by patterns of relationship aggression (Hawkins et al., 2004; Hawkins and Erickson, 2015; Rhoades and Stanley, 2011).

It is important to point out that study design appears to be an important methodological factor that requires careful consideration when interpreting the effectiveness of RE programs in addressing relationship aggression and negative aspects of relationship functioning – namely – conflict behavior. Consistent with previous meta-analyses (Hawkins et al., 2004; Hawkins and Erickson, 2015), we found that one-group-pre-post studies produced larger effect sizes than RCTs for physical and psychological aggression and conflict behaviors. In contrast, RCT study designs revealed no significant effects for either form of abuse (there were not enough studies to estimate an effect for conflict behaviors or controlling behaviors).

What are the possible reasons for these differences in effect size by way of study design? One possible explanation is that pre-post designs reveal spontaneous recovery over time. That is, change in program participants can occur progressively over time, even in the absence of the intervention (i.e., a maturational effect, Campbell and Stanley, 1966; Shadish, Cook, and Campbell, 2002). For instance, individuals or couples may be especially motivated to enrol into RE programs during periods when relationship issues are more problematic or severe, such as when conflict escalates to relationship aggression. In such instances, lower aggression at post-intervention may not reflect change due to the program, but rather naturally occurring reductions over the time frame of the study—change that would be observed in both intervention and control groups in an RCT. However, to possibly confirm or rule out this explanation, we compared the effect size (i.e., the pre-post change) of pre-post studies to that of the control groups in RCTs. We found hardly any changes in relationship aggression in the control groups of RCTs. In contrast, the effect size in pre-post studies (where all individuals/couples participated in RE programs) was 4.5 times larger and statistically significant. This analysis at least provides some evidence to suggest that pre-post study effect sizes are likely to be more than just maturational effects.

Another reason for the differences found between study designs may be due to the inadvertent yoking of participant baseline levels of relationship aggression across the two study designs. Upon close inspection, we identified that over 50% of the pre-post studies included in our review comprised of participants (either the whole sample, or a subgroup of the sample) that indexed moderate to severe relationship aggression prior to participating an RE program. This contrasts with the RCT studies in which 0.05% included participants with baseline aggression in the moderate to severe range. Unfortunately, the lack of RCT studies that included participants who indexed moderate-severe baseline aggression precluded us from conducting subgroup analysis to formally test this possible explanation. However, given that our findings suggest that RE programs are more effective for those who index greater aggression at baseline, it may be that the modest effects evidenced by RCTs may well relate to the fact that participants have pre-existing low levels of relationship aggression. The implication is that it may not be that RCTs designs suggest RE programs are less effective, rather, these studies evidence floor effects.

According to Hawkins and Erickson (2015), another possible explanation for the systematic differences in the outcomes of RE programs for the two different study designs is that a very many RCTs include much longer follow-up assessment periods (often 12-to-30-month follow-up) compared to pre-post study designs that typically conduct a single post program assessment, usually immediately following program completion. Although past RE meta-analyses do not provide much evidence for the attenuation of RE effectiveness over time (e.g., Blanchard, Hawkins, Baldwin, and Fawcett, 2009; Fawcett, Hawkins, Blanchard, and Carroll, 2010; Hawkins et al., 2004, 2012; Pinquart and Teubert, 2010), our subgroup analysis of follow-up assessments suggests that such an effect may exist for specific negative outcomes such as psychological abuse. Another possible explanation noted by Hawkins and Erickson (2015) relates to how effects are calculated in pre-post and RCT studies. Pre-post study analyses are based on treatment-on-the-treated. That is, estimates are based on those who participated in the program to some extent and completed the follow-up assessment. In contrast, the majority of RCT analyses are based on intent-to-treat. Thus, RCT estimates include responses of those assigned to undertake the program but who did not participate. Therefore, the estimates for these two study designs differ in an important way. RCT estimates reflect the expected impact of a program for an interested population, whereas pre-post study estimates reflect the effect on a program for a population that has engaged with the intervention. It is important to note that these two different estimates provide important information for decision-making regarding policy and service delivery of RE programs. On the one hand, the estimates derived in pre-post studies speak to the gains experienced by those who invest the time and effort to work through the program. On the other hand, the estimates derived from RCTs suggest that significant work is required in the future to enhance program engagement and completion for those who may be interested, but, either lack the motivation, time, or accessibility to commence RE and/or maintain participation.

6.1. Limitations and future directions

Despite the important insights gained from the current meta-analysis there are limitations that need to be taken into account. Although our subgroup analyses into methodological characteristics of RE program evaluation studies was extensive, we were still unable to conduct a number of subgroup comparisons due to the small number of studies for specific outcomes. For example, an important question remains regarding the impacts of RE programs on controlling behaviors given it is a widely acknowledged form on partner abuse (e.g., Johnson, 2008). Moreover, some of our subgroups contained the minimum requirement for the determination of effect sizes ($k = 2$) such as the testing of study design differences for specific outcomes. However, all reported subgroup analyses were powered at ≥ 0.80. Nevertheless, given the small pool of studies for some subgroup analyses (e.g., gender, dose response, program) points to the need for future research into RE programs within the context of relationship aggression.

Another limitation was that our assessment of the risk of bias for non-RCT studies identified that they had a moderate risk of bias, with one study assessed as at serious risk of bias (i.e., Rey-Anacona, Martínez-Gómez, Villate-Hernández, González-Blanco, and Cárdenas-Vallejo, 2014). Although conducting a quantitative synthesis that combines studies with low to moderate bias is regarded as acceptable, caution must be exercised when including and interpreting the outcomes of studies assessed as having a serious risk of bias. In our meta-analysis we chose to include the Rey-Anacona et al. (2014) study to ensure we the captured the full breadth of studies conducted in the field (given the field is somewhat limited), but also, examined the homogeneity of this study (and all other non-RCTs as well as the included RCTs) across critical study aspects (i.e., intervention being trialed, sampling population, outcome measures, and alike). Our examination noted high degree of homogeneity across these study design features. Furthermore, subgroup analyses and sensitivity analyses were conducted to detect various study characteristics that may account for differences in effect sizes.

A final important limitation relates to the inability of our meta-analysis to disentangle certain sample characteristics with outcomes such as baseline levels of relationship aggression and socio-economic status. This is because, although all included studies provided baseline assessments of relationship aggression, and we were able to group studies into varying levels of relationship aggression, the majority of studies involved low-income participants. Thus, there is some ambiguity as to the whether the effects of RE programs clearly reflect outcomes for those who vary in relationship aggression, experience social disadvantage, or both.

The limitations we have outlined speak to the boundary conditions of the current meta-analysis. Therefore, the potential policy implications of our findings should be considered in light of the need to further understand how particular sample characteristics and study design features are important to consider in the implementation of RE programs to address relationship aggression.

6.2. Conclusion

The findings of this meta-analysis provide important insights into the potential for RE programs to address issues of relationship aggression and to reduce negative relationship functioning of those with prior experiences of relationship aggression. Although RE programs were not originally developed to tackle issues of relationship aggression, the education and skills that individuals and couples can derive from such programs should mitigate destructive relationship behaviors that constitute relationship aggression. Indeed, the findings of this quantitative synthesis highlight that RE programs show promise in reducing the experience of relationship aggression as well as attenuating conflict behaviors. Importantly, these effects are larger in samples of individuals who experience higher incidents of relationship aggression prior to entering RE programs. However, research into the application of RE programs for those who experience relationship aggression is still emerging, and it is critical that a larger evidence-base is amassed before there are clear indications that RE programs are indeed effective for those experiencing aggression within their romantic relationships. Nevertheless, the promising findings to date highlight the potential of RE programs to address this important societal issue.

Declaration of Competing Interest

None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cpr.2023.102285.

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